

HRP-203-Assignment-4

Introduction

Smoking is the leading preventable cause of cardiovascular disease, responsible for about 1 in 4 cardiovascular disease-related deaths in the United States. It raises triglycerides, lowers "good" cholesterol (HDL), increases the tendency of blood to clot-which can block blood flow to the heart and brain, damages cells that line the blood vessels, increases the buildup of plaque in blood vessels, and causes thickening and narrowing of blood vessels. Smoking increases the risk of many cardiovascular diseases, including atherosclerosis, coronary heart disease (CHD)-including heart attack and sudden cardiac death, stroke, peripheral arterial disease (PAD), and abdominal aortic aneurysm. ([CDC, 2024](#))

Smoking also has a dramatic effect on the economy and the healthcare sector. In 2018, cigarette smoking cost the United States more than an estimated \$600 billion, including more than \$240 billion in health care spending. ([CDC, 2024](#)) In a cross-sectional study of over 13,000 adults with chronic lower respiratory disease, the annual smoking-attributable health care expenditures (SAHEs) was found to be \$13.6 billion, or \$2,752 per current smoker and \$1,083 per former smoker for people ages 35 to 65. Among those 65 years or older, annual SAHEs were \$5.3 billion, or \$1,704 per current smoker and \$682 per former smoker. ([Gu et al., 2024](#))

Methods

A simulated cohort (n = 5000) dataset was analyzed for the purposes of this study. Two generalized linear models were fit. A logistic regression (binomial family) estimated the association between smoking and cardiovascular disease incidence. A linear regression (gaussian family) assessed the association between smoking and healthcare costs. Both models adjusted for age and sex. Below are the model specifications:

$$\log(p / (1-p)) = \beta_0 + \beta_1(\text{smoke}) + \beta_2(\text{female}) + \beta_3(\text{age})$$

where p is the probability of cardiovascular disease incidence

$$E[\text{cost}] = \beta_0 + \beta_1(\text{smoke}) + \beta_2(\text{female}) + \beta_3(\text{age})$$

To validate the models, a 70/30 train-test data split was created. The training dataset was used to build the regression models, and the test dataset was used to compare predicted outcomes to actual outcomes. Validation was plotted as an ROC curve for the logistic regression (smoking and cardiovascular disease incidence) and as a plot for the linear regression (smoking and healthcare costs).

The models were then fit to the full dataset to assess the direction and magnitude of association between smoking and each outcome: cardiovascular disease incidence and healthcare costs.

Results

Table 1 describes the simulated cohort (n = 5,000). The average age was 45 (95% CI: 31–59), and 2,866 (57%) were female. Of the 5,000 individuals, 646 (13%) were smokers, 275 (5.5%) had been diagnosed with cardiovascular disease, and the average healthcare cost was \$9,376 (95% CI: \$9,090–\$9,676).

Characteristic	N = 5,000 ¹
smoke	646 (13%)
female	2,866 (57%)
age	45 (31, 59)
cardiac	275 (5.5%)
cost	9,376 (9,090, 9,676)
row_id	2,501 (1,251, 3,751)
¹ n (%); Median (Q1, Q3)	

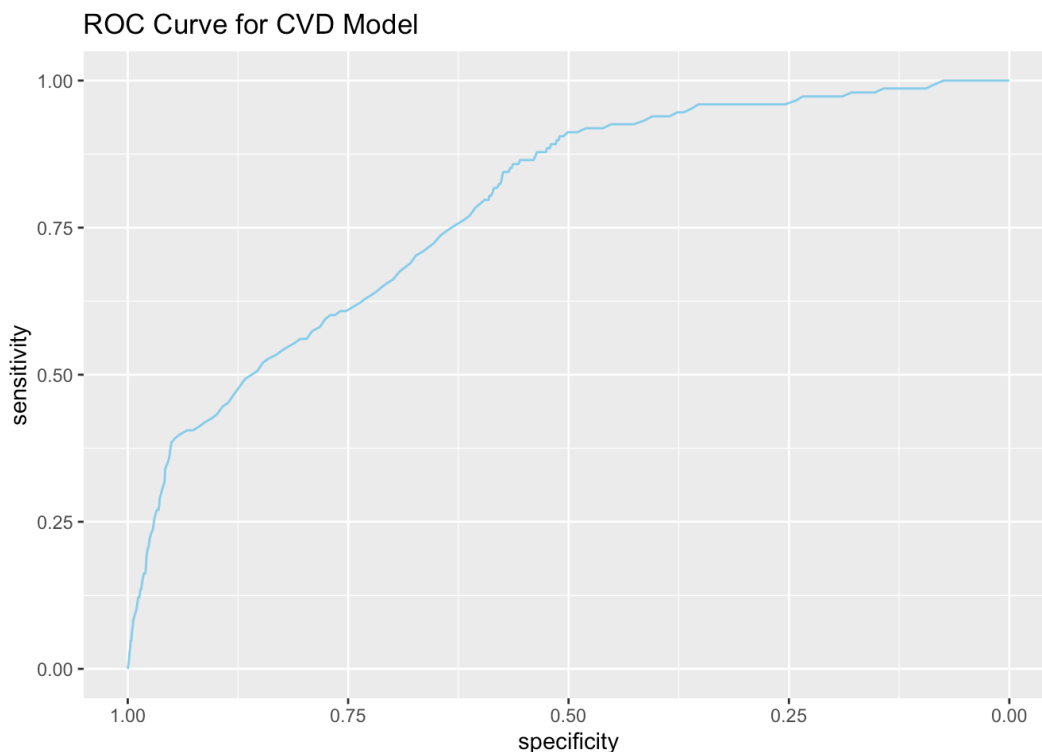
Table 1: Descriptive statistics of the simulated cohort (n = 5,000), including smoking status, cardiovascular disease incidence, healthcare costs, age, and sex.

Using a base logistic regression model to assess the relationship between smoking and cardiovascular disease incidence on the training dataset, not adjusting for age and sex, smokers have $\exp(2.14) = 8.47$ times the odds

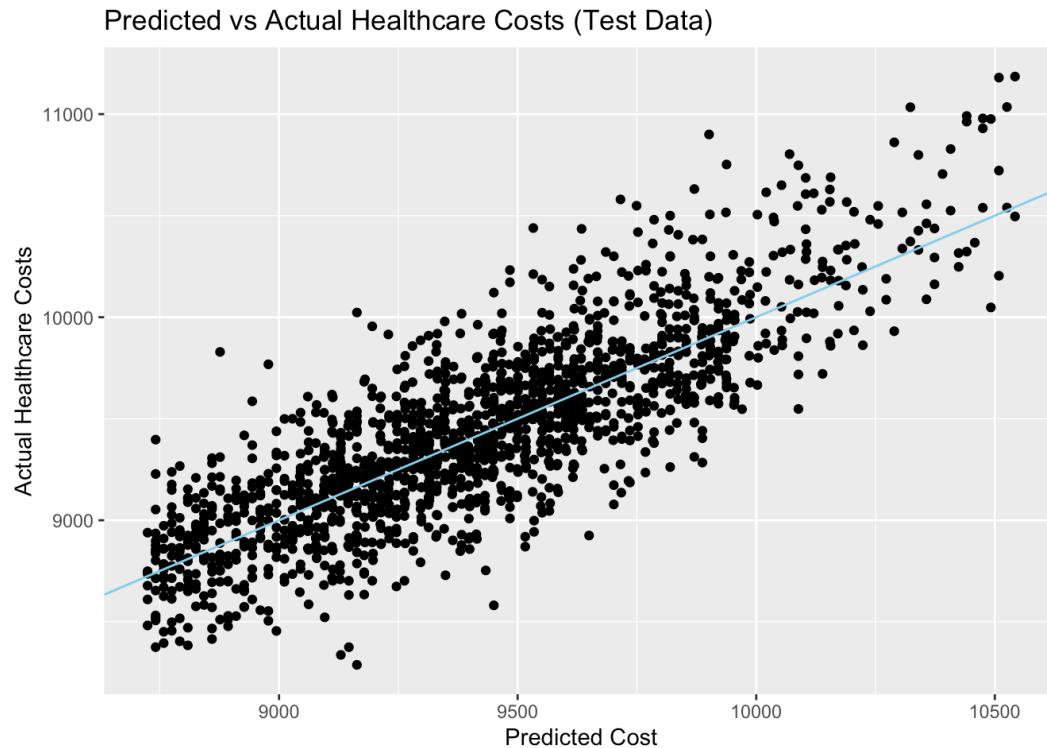
of being diagnosed with cardiovascular disease compared to non-smokers ($p < 0.001$). Adjusting for age and sex, smokers have $\exp(2.50) = 12.21$ times the odds of being diagnosed with cardiovascular disease compared to non-smokers ($p < 0.001$). Female sex was associated with substantially lower odds of CVD ($\exp(-4.38) = 0.01$, $p < 0.001$). Age was not statistically significant at the 0.05 level ($\exp(0.011) = 1.01$, $p = 0.077$).

Using a base linear regression model to assess the relationship between smoking and healthcare costs on the training dataset, not adjusting for age and sex, smokers face an average increase of \$552.38 in healthcare costs compared to non-smokers ($p < 0.001$). Adjusting for age and sex, smokers face an average increase of \$587.69 in healthcare costs compared to non-smokers ($p < 0.001$). Female sex was associated with \$318.94 lower healthcare costs ($p < 0.001$). Each additional year of age was associated with \$16.87 higher healthcare costs ($p < 0.001$).

We then tested the model fit on the test data. Figures 1a-b demonstrates the results of the validation. We see that the Receiver Operating Characteristic (ROC) curve yielded an area under the curve (AUC) of approximately 0.78, indicating some discriminative ability of the cardiovascular disease model. The validation plot for the cost model showed reasonable agreement between predicted and actual values.



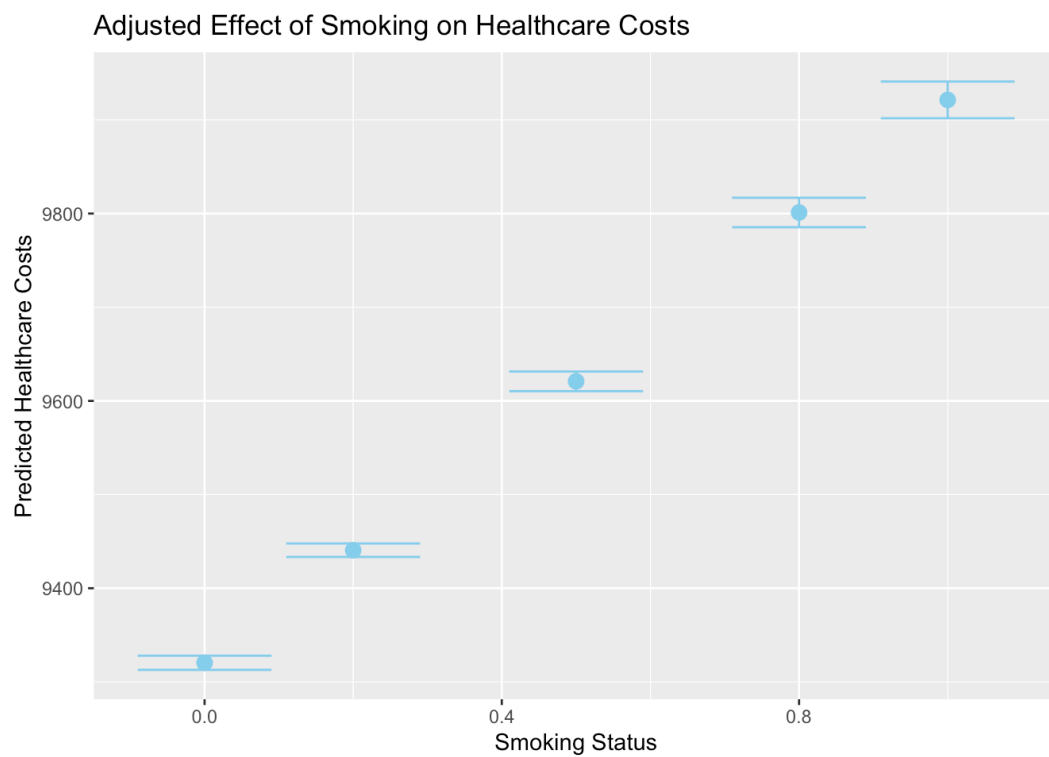
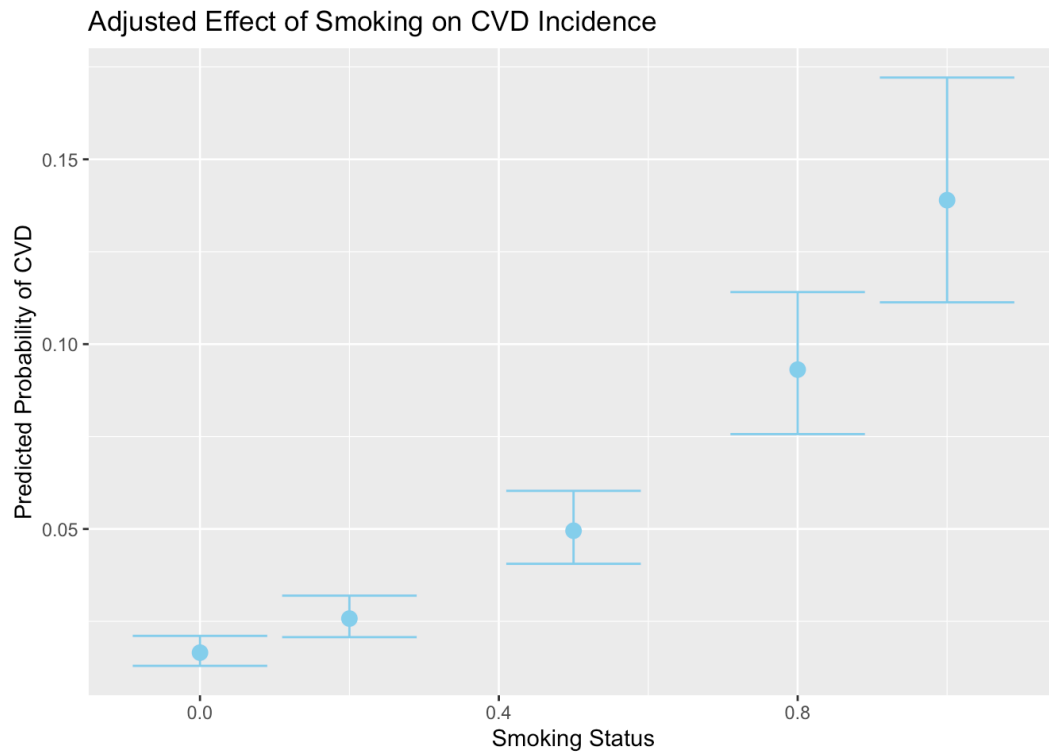
Area under the curve: 0.7836



Figures 1a-b: Model validation on the test dataset (30% holdout). (a) ROC curve for the logistic regression model predicting cardiovascular disease incidence (AUC = 0.784). (b) Predicted vs. actual healthcare costs from the linear regression model; the diagonal line represents perfect agreement.

Fitting the logistic regression to the full dataset, adjusting for age and sex, smokers have $\exp(2.26) = 9.58$ times the odds of being diagnosed with cardiovascular disease compared to non-smokers ($p < 0.001$). Female sex was associated with substantially lower odds of CVD ($\exp(-2.56) = 0.08$, or approximately 92% lower odds, $p < 0.001$). All coefficients were statistically significant at $p < 0.05$. Age was marginally significant in the training data ($p = 0.077$) but reached significance when fit to the full dataset ($p = 0.031$), likely reflecting increased statistical power.

Fitting the linear regression to the full dataset, adjusting for age and sex, smokers face an average increase of \$600.87 in healthcare costs compared to non-smokers ($p < 0.001$). Female sex was associated with \$323.76 lower healthcare costs ($p < 0.001$). Each additional year of age was associated with \$16.96 higher healthcare costs ($p < 0.001$). All coefficients were statistically significant at $p < 0.05$.



Figures 2a-b: Adjusted associations from models fit to the full dataset, controlling for age and sex. (a) Predicted probability of cardiovascular disease by smoking status with 95% confidence intervals. (b) Predicted healthcare costs by smoking status with 95% confidence intervals.

Discussion

Overall, we found a strong positive association between smoking and both outcomes. Smokers had 9.58 times the odds of CVD diagnosis compared to non-smokers, adjusting for age and sex ($p < 0.001$), and incurred an estimated \$601 more in annual healthcare costs ($p < 0.001$). These findings are consistent with the trends reported in the literature on smoking-attributable morbidity and healthcare expenditures.

Among the covariates, female sex was associated with substantially lower odds of CVD ($OR = 0.08$, $p < 0.001$) and lower healthcare costs ($-\$324$, $p < 0.001$), consistent with known sex differences in cardiovascular risk. Age was a modest yet significant predictor of both outcomes.

There are several limitations of note. First, this analysis uses simulated data, so the effect sizes may not generalize to real-world populations. Second, the models adjust only for age and sex; unmeasured confounders such as diet, physical activity, socioeconomic status, and comorbid conditions are likely important predictors of both cardiovascular disease and healthcare costs that could bias our estimates. Third, the logistic regression model achieved an AUC of 0.78, indicating moderate discriminative ability—additional predictors would likely improve classification. Finally, the cross-sectional design of this study precludes causal inference; longitudinal data would be needed to establish a temporal relationship between smoking exposure and outcomes.

AI Statement

I attest to the fact that I did not use generative AI technology to complete any portion of this work.